How to Build a Time Series Agent

Reuse what is general · adapt what is temporal · rebuild what is task-bound

ALIGN · LLM

interprets temporal evidence and selects actions

- Represent numbers for the language model
- Reason over temporal evidence
- Learn which workflow action to take

→ Not enough alone: it acts through the harness

REUSE · General Harness

domain-independent runtime

- Interaction & memory
- Tools & execution
- Planning & coordination
- Feedback & self-improvement

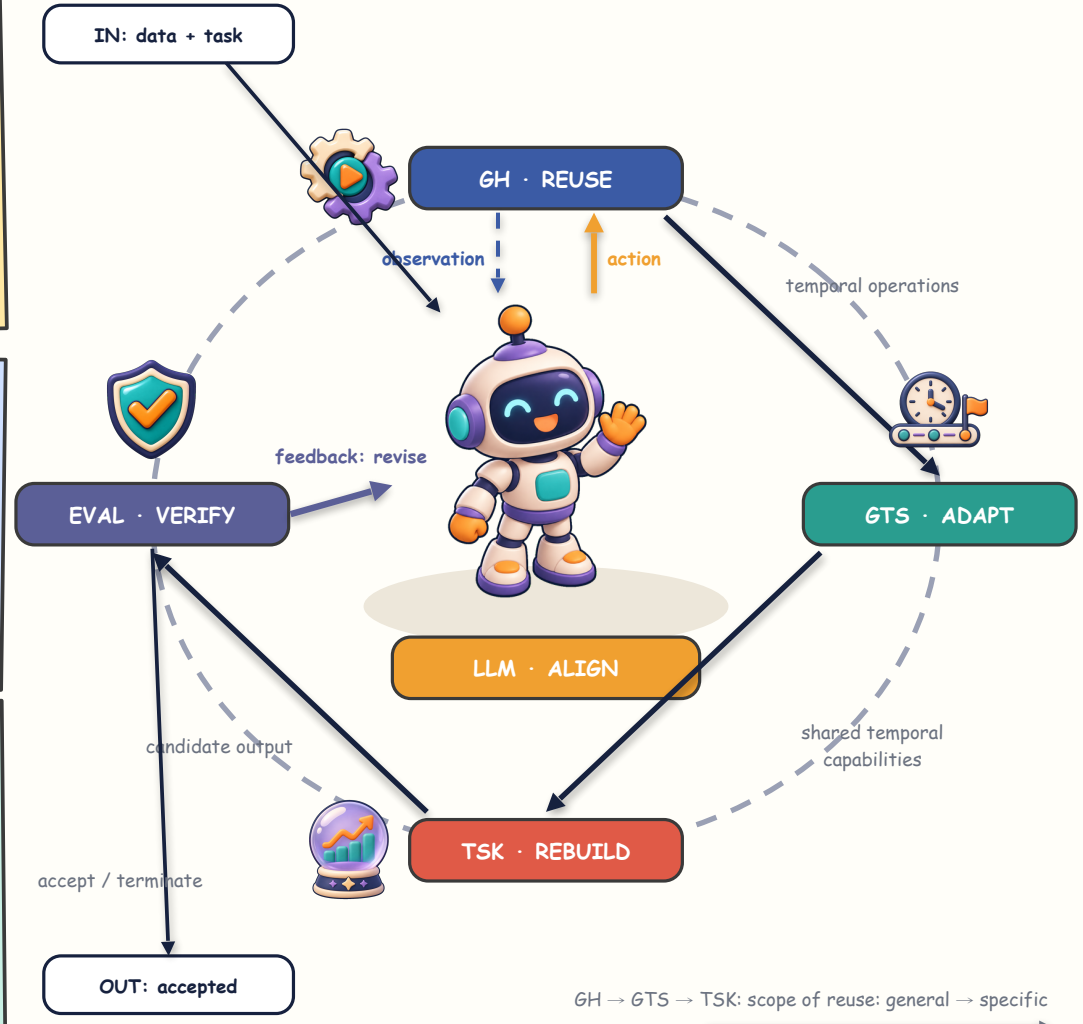
→ Carry it over from general Agents

ADAPT · General Time Series Harness

temporal semantics shared across tasks

- Timestamps, units, missingness
- Windows, resampling, decomposition
- Only information available at t_0
- Chronology guards & validators

→ Add time where a generic runtime is ambiguous



REBUILD · Task-Specific Harness

procedures bound to one task contract

- Forecasting
- Synthesis & augmentation
- Anomaly diagnosis
- Decision support

→ Rebuild around output, objective and actions

VERIFY · Evaluation

tests the complete workflow

- Process validity
- Temporal validity
- Task performance
- Decision utility

→ Valid run ≠ accurate result ≠ useful decision

IN Scenario & data
series · context · task request

OUT Accepted output
forecast · diagnosis · decision

KEY CLAIM
Align model capability, harness responsibility and evaluation